Research Plan (Paper 1): Closing the Frequency Gap

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***Abstract*—Paper 1 targets ICRA 2027 (PaperPlaza deadline: 15 Sep 2026 PST): *Closing the Frequency Gap: Real-Time Humanoid Control via Vision-Language-Action Models and Echo State Networks.* Primary author: Andrew. Co-author: Tariq. This revision (5 Aug 2026) sanity-checks measured progress, sharpens motivation, locks a 6-week path to submission, and adds an onboarding primer (symbols, metrics, plain-language insights).**

I. Onboarding Primer (Read This First)

This section is for collaborators joining mid-project. Skim it once; then the status numbers and equations will make sense.

A. The project in one paragraph

A Vision-Language-Action model (**VLA**, here Unitree **UnifoLM**) looks at the camera + a language goal and outputs desired joint angles for the Unitree **G1** humanoid (29 joints). That VLA is smart but slow ($\sim$1.7 updates per second). The robot needs smooth commands at **100 Hz** (100 times per second). We insert an **Echo State Network (ESN)** between them: the VLA gives sparse targets; the ESN fills in smooth 100 Hz joint commands. Baselines that must lose to us (or we have no paper): hold the last VLA output (**ZOH**) or linearly interpolate between VLA outputs.

B. Symbols and knobs (what α, λ, . . . mean)

Symbol	Name	Plain meaning
α	Leak rate	How fast the ESN forgets. Update mixes old state and new input: $(1-\alpha)$ old + α new. Our best: α=0.3 (keep $\sim$70% memory, take $\sim$30% new). Smaller $\alpha \Rightarrow$ longer memory / smoother; larger $\alpha \Rightarrow$ more reactive.
λ	Ridge strength	Penalty on large readout weights when fitting W_{out}. Our best: λ=1 (fairly strong; prefers stable fits). Tiny λ overfits demos and can explode error.
N	Reservoir size	Number of hidden units (we use N=1000). Bigger $\Rightarrow$ more capacity and compute.
ρ	Spectral radius	Scales internal recurrence of W_{res} (we use ρ=0.95). Keeps the "echo" rich without blowing up.
$u(t) \in \mathbb{R}^{58}$	ESN input	$[q_t;\, q^{\star}_{\text{VLA}}]$: current 29 joints + latest 29-D VLA target.
$y(t) \in \mathbb{R}^{29}$	ESN output	Next 100 Hz joint command sent toward the robot / sim.
$x(t)$	Reservoir state	Internal N-D memory; not trained by backprop.
W_{out}	Readout	Only trained part (ridge regression). Reservoir weights stay frozen.

Core ESN update (leaky reservoir):

$$x(t) = (1-\alpha)x(t-1) + \alpha\,\tanh\big(W_{\text{res}}x(t-1) + W_{\text{in}}u(t)\big),$$
$$y(t) = W_{\text{out}}[x(t); u(t)].$$

C. Metrics you will see in logs

- **Latency (ms) / Hz:** How long one VLA inference takes; Hz = 1000/ms. Gap vs. 100 Hz $\approx$ 100/Hz (e.g., 1.745 Hz $\Rightarrow\sim$57$\times$ too slow).
- **RMSE (rad):** Average joint-angle error. 9.83×10^{-4} rad $\approx 0.056°$—very tight offline tracking.
- **Jerk / jerk ratio:** Smoothness (derivative of acceleration). Ratio $\approx$0.981 means predicted motion is nearly as smooth as the demo (1.0 = matched).
- **P99 / steady max (ms):** Tail control-step time. Budget is 10 ms for 100 Hz; we are under that when VLA is mocked.
- **Mock VLA vs. live UnifoLM:** Mock = fake fast tokens (timing test only). Live = real model (needed for the paper claim).
- **Oracle / kinematic replay:** Drive the ESN with recorded demo signals in MuJoCo—tests bridge fidelity, *not* closed-loop policy success.

D. Acronym cheat sheet

VLA vision-language-action model; **ESN** echo state network (reservoir computer); **ZOH** zero-order hold; **S2R** sim-to-real; **DIRT** domain / discrepancy transfer tooling in our stack; **GIL** Python global interpreter lock (we bypass it with multiprocessing); **DoF** degrees of freedom (G1: 29); **MuJoCo** physics simulator.

E. Plain-language insights (current "Solid" evidence)

1) **Frequency gap is real.** Live UnifoLM on our DGX is $\sim$573 ms per action ($\sim$1.7 Hz). Two profilers agree; nothing failed to parse. The robot wants 100 Hz—so naive hold looks like a staircase of joint targets.
2) **ESN can track offline.** With α=0.3, λ=1, a frozen reservoir reproduces 100 Hz joints from sparse VLA holds almost as well as the demo (tiny RMSE, jerk matched). The *bridge idea works on recorded data.*
3) **The controller is fast enough.** With a fake VLA, MuJoCo+ESN runs at $\sim$248 Hz. The ESN loop is not the bottleneck; the slow VLA is.
4) **S2R code exists, results do not yet.** Deploy configs and ablation YAMLs are real; measured hardware transfer metrics are still future work.
5) **What this does *not* prove.** We have not yet shown that live UnifoLM $\rightarrow$ ESN beats ZOH/linear on wipe

success. That comparative closed-loop study is the remaining core of the paper.

II. Sanity Check: What Is Actually Done

Source of truth: measured artifacts under `research/results/` and the draft in `papers/icra2027/main.tex` (status date on manuscript: 30 Jul 2026). Today: **5 Aug 2026** ⇒ ∼**6 weeks** to deadline—not a 12-week runway.

A. Solid (publishable as supporting evidence)

- **Frequency gap (live UnifoLM, not mock).** PyTorch n=100 on `G1_Clean_Table`: mean **573.2 ms / 1.745 Hz** (gap 57.3× vs. 100 Hz); Nsight NVTX mean **593.7 ms / 1.684 Hz**. Dual instruments agree; parse failure rate 0%. This is the strongest empirical claim so far.
- **CUDA ESN + ridge readout (offline).** Sweep on wipe ep. 0 selects $(\alpha, \lambda) = (0.3, 1)$: joint RMSE 9.83×10^{-4} **rad**, jerk ratio ≈0.981 vs. demo. Training cheap (∼2.9 kHz). Claim: *a frozen reservoir can track 100 Hz joints from sparse VLA holds.*
- **Control timing headroom (mock VLA).** GIL-free dual-process MuJoCo+ESN sustains **248 Hz** mean (P99 5.42 ms; steady max 6.82 ms <10 ms). Claim: *the bridge clears the 100 Hz budget when VLA is not the bottleneck.*
- **S2R scaffolding.** Deploy package, ablation YAMLs (ZOH / raw / ESN), G1 29-DoF defaults exist. Not yet a result—but the transfer path is real code, not vapor.

B. Overclaimed or Misframed in Prior Plan

- **"MuJoCo eval complete / 17% grasp" is wrong.** The measured run is *kinematic oracle replay* (dataset proprio + held VLA tokens through the trained ESN), not live UnifoLM closed-loop. `grasp_success=true` on *one* episode; 1047/6000 frames attached ≈17% of *frames*, not a 17% success rate. Joint RMSE 1.37×10^{-3} rad is tight; right-EE RMSE ∼9.6 cm and table-contact 0.38% show geometric wipe quality is still weak.
- **"Dual-process complete" ≠ closed-loop complete.** Report flag: `mock_vla: true`. Timing is done; policy evidence is not.
- **Ablations / baselines / paper figures marked ready elsewhere are empty.** `results/step1_baselines`, `step3_ablation`, `step3_evaluation`, `step4_paper_*` are README placeholders only. Code and configs exist; comparative tables do not.
- **Page budget.** ICRA contributed papers: **6 pages + 1 optional refs page**, not 8-page freestyle.
- **Timeline inconsistency.** Prior plan mixed a 12-week calendar with a Sep 15 submit week ending in October. Impossible. Rewrite below is deadline-aligned.

C. Honest Status Table

Bottom line: We have a credible *systems + offline dynamics* story. We do **not** yet have the *control* result that ICRA reviewers will demand: live UnifoLM → ESN → task success vs. ZOH/linear on the same protocol.

TABLE I
Milestone audit (5 Aug 2026). Green = measured artifact; amber = code/config only; red = missing for submission.

Milestone	**Status**	**Supports**
UnifoLM latency (PyTorch+Nsight)	Done	Frequency-gap claim
ESN ridge train + $\alpha \times \lambda$ sweep	Done	Offline tracking claim
Dual-process @>100 Hz (mock VLA)	Done	Timing / systems claim
MuJoCo wipe oracle (1 ep., kinematic)	Done*	Bridge fidelity, not policy
Live UnifoLM closed-loop wipe	Missing	**Core paper claim**
Baselines: ZOH / linear / ESN	Code only	Comparative tables
N/ρ ESN ablations (closed-loop)	Partial	Offline only; need trials
S2R/DIRT transfer metrics on G1	Scaffold	Sim-to-real paragraph
Paper figures + baseline tables	Empty	Camera-ready results
IEEE draft skeleton	Draft	Needs live numbers

III. Improved Motivation

A. Problem (reframed)

Foundation VLAs (OpenVLA, π_0, UnifoLM-VLA) make open-vocabulary manipulation plausible on humanoids, but they are *slow policies in a fast plant*. On our hardware, UnifoLM-VLA-Base updates joint intents at ∼1.7 Hz while the Unitree G1 Edu expects 100 Hz commands over 29 DoF. Under naive zero-order hold, each VLA token is held for ∼57 control ticks: the robot sees a staircase of targets, not a trajectory. That mismatch is not a minor engineering annoyance—it is a structural incompatibility between foundation-model inference and embodied control rates.

B. Why this is a research question (not just a filter)

1) **Rate mismatch is first-class.** Speeding up the VLA (quantization, distillation, smaller heads) shrinks the gap but does not remove the need for a continuous dynamical interface while the next token is computing. We treat the bridge as a scientific object: what temporal structure must sit between sparse language-conditioned intents and dense joint actuation?
2) **Simple upsamplers are inadequate hypotheses.** ZOH preserves discontinuities; linear interpolation ignores plant dynamics and contact. Both are necessary baselines. If ESN fails to beat them on success and jerk under live VLA, the hybrid claim collapses—and that falsifiability is a feature.
3) **Why ESN specifically.** Reservoir computing gives (a) millisecond forward passes with no BPTT, (b) a frozen nonlinear memory matched to continuous-time motor streams, and (c) a one-shot ridge readout that can be retuned from short demos. Unlike replacing the VLA with a faster policy, we keep language grounding and add a reflexive temporal layer. Unlike CPGs, the drive is language- and vision-conditioned, not purely rhythmic.

C. Core claim (one sentence)

A leaky Echo State Network, driven by concatenated proprioception and sparse VLA joint targets $u(t) = [q_t; q^{\star}_{\mathrm{VLA}}] \in \mathbb{R}^{58}$, can close the measured ∼57–59× frequency gap and improve task success and motion smoothness over ZOH and linear upsampling on G1 wipe/clean tasks—without retraining the VLA.

D. Architecture (unchanged; clarified role)

Camera $\rightarrow$ UnifoLM-VLA ($\sim$1.7 Hz measured) $\rightarrow$ ESN bridge (100 Hz) $\rightarrow$ G1 joints (29 DoF)

- **VLA:** open-vocabulary intents (slow, semantic).
- **ESN:** temporal continuity and rate matching (fast, reflexive).
- **Plant:** G1 Edu / MuJoCo stand-in until live closed-loop is green.

IV. Research Trajectory (Evidence Ladder)

Each rung answers a distinct reviewer question. Do not skip rungs; do not oversell lower rungs as higher ones.

1) **Rung A — Gap exists (DONE).** Q: Is UnifoLM too slow for 100 Hz G1 control?
   A: Yes—573 ms / 1.745 Hz (PyTorch), 594 ms / 1.684 Hz (Nsight).
2) **Rung B — Bridge can track offline (DONE).** Q: Can a ridge ESN reproduce 100 Hz joints from sparse holds?
   A: Yes—9.83×10^{-4} rad RMSE at $(\alpha, \lambda) = (0.3, 1)$.
3) **Rung C — Bridge is fast enough in-loop (DONE, mock).** Q: Does the control process clear 10 ms?
   A: Yes—248 Hz mean under mock VLA.
4) **Rung D — Bridge preserves task geometry in sim (PARTIAL).** Q: Does ESN replay yield grasp/wipe behavior in MuJoCo?
   A: Joint tracking yes; wipe contact / EE accuracy still weak; **only oracle, 1 episode**. Next: multi-episode oracle + live-VLA sim (no mock) before hardware.
5) **Rung E — Live UnifoLM closed-loop beats baselines (CRITICAL / MISSING).** Q: On the same wipe protocol, does VLA+ESN beat ZOH and linear on success, jerk, and latency?
   A: TBD. This is the paper's make-or-break experiment.
6) **Rung F — Ablations explain *why* (MISSING TABLES).** Sweep $N \in \{500, 1000, 2000\}$, $\rho \in \{0.85, 0.95, 1.05\}$, and α around 0.3; report RMSE, jerk, success.
7) **Rung G — Transfer note (OPTIONAL IF TIME).** S2R/DIRT metrics on G1 strengthen the systems story; if blocked, keep as limitation + deploy description (honest).

Paper narrative = Rungs A–C as setup, D as sanity, E–F as results. Without E, the manuscript is a profiling + offline ESN note—not an ICRA control paper.

V. Critical Path (6 Weeks to 15 Sep 2026)

A. Phase 1 — Replace mock with live UnifoLM in sim (Week of Aug 5–14)

Goal: Same dual-process loop, `--mock` off; log success / Hz / p99 / jerk.

- Run `step3_dual_thread_mujoco` live; freeze episode protocol (wipe primary; clean/grasp secondary if time).
- Implement/verify baselines on identical env: ZOH, linear interpolation, ESN (checkpoint already selected).
- Deliverable: filled `results/step3_*` JSON + first comparison table (even if n is small).
- Advisor checkpoint target: mid-August.

B. Phase 2 — Comparative study + ablations (Aug 15–31)

Goal: Rung E–F numbers that can enter Table I / ablation table.

- Trial budget (realistic for 2–3 weeks): **$\geq$20 trials/method** on wipe in sim (ZOH / linear / ESN); scale to 50 only if runtime allows.
- Hardware G1 live trials: attempt in parallel via S2R; **do not block submission on full 150-trial hardware suite**. Prefer fewer well-logged hardware episodes over an unfinished mega-protocol.
- Ablations: N and ρ at fixed best (α, λ); optional $\alpha \in \{0.1, 0.3, 0.5\}$ confirmation under live VLA.
- Metrics (fixed set): task success (binary), joint RMSE, right-EE RMSE, jerk, control Hz / p99 latency, contact ratio (wipe).

C. Phase 3 — Manuscript freeze (Sep 1–12)

- Fill measured tables; generate trajectory overlays and bar charts from logs (no placeholder figures).
- Soften abstract/intro: every claim tagged DONE vs. TBD must match artifacts.
- Length discipline: 6 pages body + optional refs page; video $\leq$180 s.
- Tariq: technical review (methods, equations, result consistency) within 3 days of draft freeze.

D. Phase 4 — Submit (Sep 13–15)

Font-embedded PDF; PaperPlaza upload $\geq$48 h before deadline; backup venue list ready if travel/presentation constraints force a pivot.

E. Week-by-Week Owners

Window	Focus	Owner	Exit criterion
Aug 5–14	Live UnifoLM sim + baselines	Andrew lead	Non-mock step3 + 3-way table
Aug 15–22	Trial campaign (sim)	Both	$\geq$20 trials $\times$ 3 methods
Aug 23–31	Ablations + optional G1	Tariq lead	Ablation CSV + hardware logs
Sep 1–7	Figures + Results rewrite	Andrew	Plots in `papers/`
Sep 8–12	Full draft + review	Both	Reviewed 6+1 PDF
Sep 13–15	Submit	Both	PaperPlaza confirmation

VI. Collaboration Split (Unchanged Intent, Clearer Scope)

- **Andrew:** architecture, protocol design, UnifoLM/S2R integration, writing, figures narrative, submission.
- **Tariq:** baseline/ablation execution, trial logging, plot generation from logs, methods accuracy review.
- **Non-goals for Tariq this cycle:** redesigning the VLA, rewriting the paper from scratch, inventing new architectures.

VII. Risks (Reframed Around the Critical Claim)

- **Live UnifoLM closed-loop fails to beat ZOH.** Mitigation: report negative result honestly with profiling + offline ESN as systems contribution; strengthen ablations; consider shorter-horizon wipe subtasks. Do not hide behind mock numbers.
- **Hardware access slips.** Mitigation: primary results = live-VLA MuJoCo; G1 = best-effort transfer note (Rung G optional).
- **Scope creep (3 tasks $\times$ 50 trials $\times$ all ablations).** Mitigation: wipe-only primary; clean/grasp as appendix or future work; 20-trial floor before 50-trial stretch.
- **ICRA on-site presentation constraint.** Mitigation: co-author travel plan + backup venues already listed in the umbrella 2026 plan.

VIII. Immediate Next Actions (This Week — Sim)

1) Prefer notebooks (kernel: Research Summer 2026 / UnifoLM), not CLI.
2) Offline ZOH/linear: open `notebooks/step3_control_baselines.ipynb` $\rightarrow$ Run All.
3) Live UnifoLM bridges: open `notebooks/step3_dual_thread_mujoco.ipynb`, set `MOCK=False`, run with `BRIDGE="esn"`, then `"zoh"`, then `"linear"`.
4) Or one-shot: `notebooks/step3_sim_comparison.ipynb` with `MOCK=False` $\rightarrow$ Run All.
5) Next week: Jetson AGX Thor $\rightarrow$ G1 via `notebooks/step5_s2r_*.ipynb`.